

Topic 1: Spatial data systems

basic concepts

jose.moreira@ua.pt

Contents

Concepts and vocabulary

Geographic Information Systems (GIS) and Spatial DBs points of view

Modes and models of representation of spatial data

Spatial data formats

Spatial object

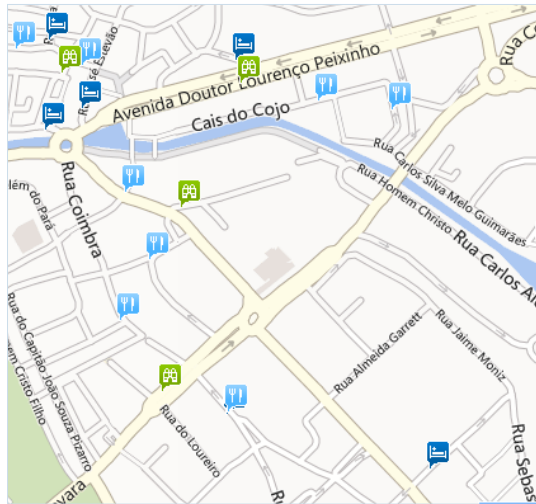
A spatial object has non-spatial and spatial attributes

- A *hotel* has *name* and *category* which do not carry spatial semantics
- The *address* of a hotel is a non-spatial attribute (string) but it is associated with spatial semantics (a location on a city map)
- **Non-spatial spatially enabled systems** (e.g., a conventional DBMS) can retrieve the hotels of a particular category
... but **cannot retrieve the nearest hotel (e.g., considering walking distance) to a given location** (e.g., exhibition center)

Spatial component

Geometry: location and shape

Topology: spatial relationships among objects, e.g., adjacency or intersection (nodes in a network)



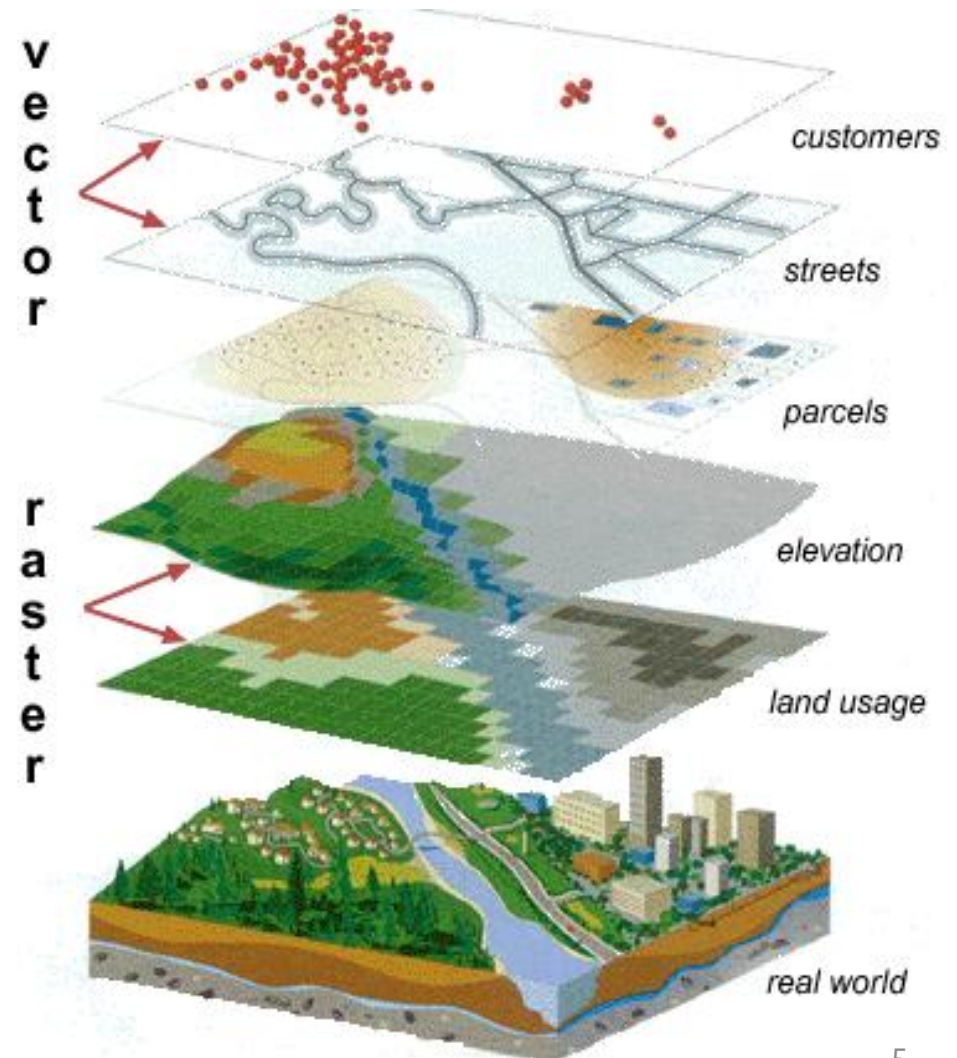
Problem: how to represent spatial components?

Spatial attributes (geometry) does not correspond to any base data type

Theme

A theme gathers geographical information regarding a particular topic

- it consists of a set of homogeneous spatial objects
- it is like a relation in a relational model



Applications

Users of mobile devices may want to find the nearest restaurant to their location

Astrologers may want to find spatial relationships among objects in the universe

Agriculture authorities may want to study soils occupation

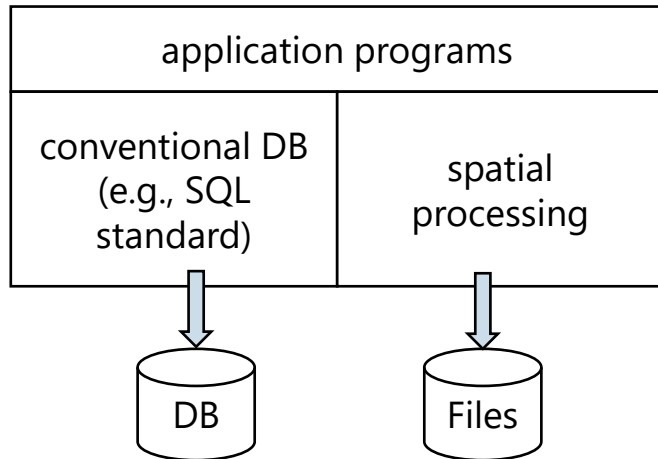
Environmentalists may want to develop models of sustainable forest harvesting

National authorities may want to find the best location for an airport

GIS vs Spatial DBMS

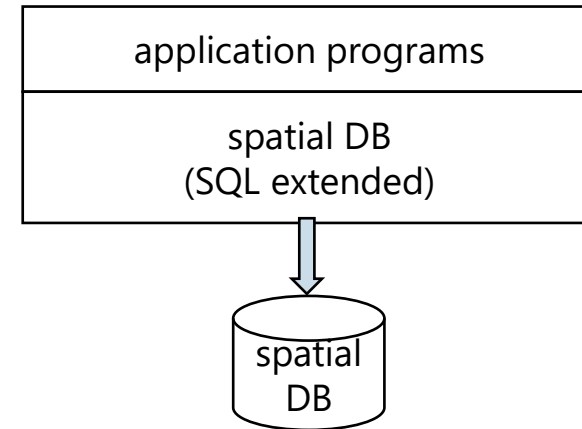
GIS

- Loosely coupled approach
- The goal is to assist the **analysis** and **visualization** of spatial data



Spatial DBMSs

- Integrated approach
- Extends a conventional DBMS to support spatial data



GIS vs Spatial DBMS

GIS

- The coexistence of heterogeneous data models makes difficult the modeling and integration of non-spatial and spatial data
- Partial loss of basic DBMS functionality, such as transaction management, recovery and optimization
- GIS can be built on the top of a spatial DBMSs

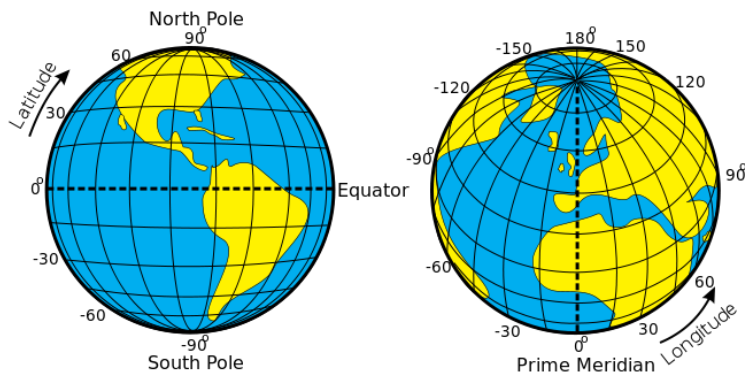
Spatial DBMSs

- The SQL language is extended to manipulate non-spatial as well as spatial data
- Other DBMS functions such as query optimization are upgraded to handle spatial data efficiently
- Not exclusively for geographical content; there are spatial objects in medical images, computer-aided design (CAD), stars in the sky...

Geographic coordinate systems

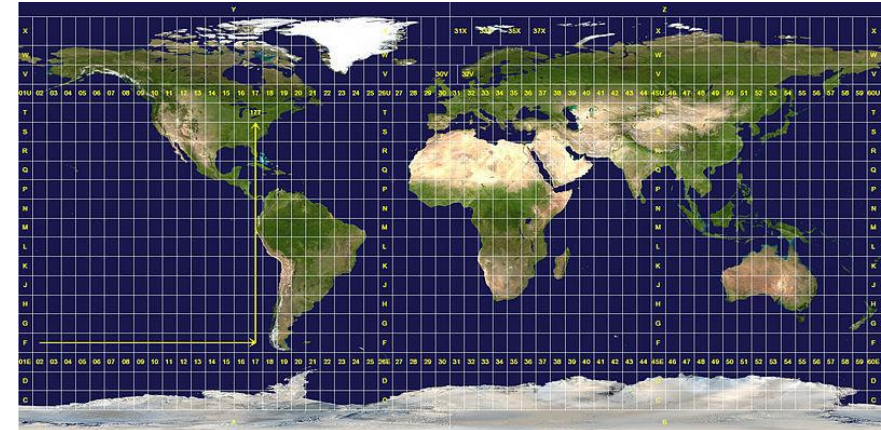
Geographic

- latitude, longitude



Universal Transverse Mercator (UTM)

- 60 UTM zones in longitude
- 20 zone designator in latitude



Each zone (row) has a scale factor
Low distortion ($\sim 1/1000$)

Datums

The Earth is not a perfect sphere

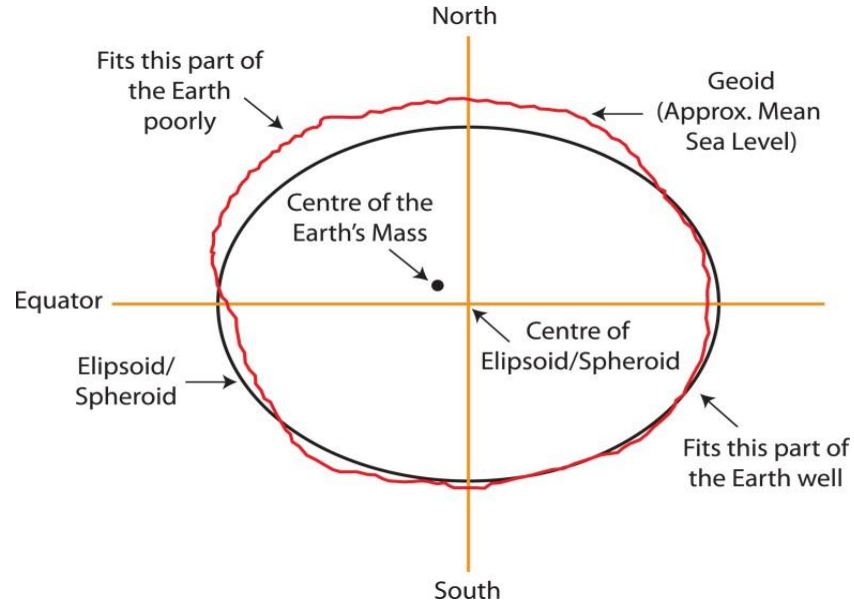
The datum defines the spheroid that is used

Datums can be

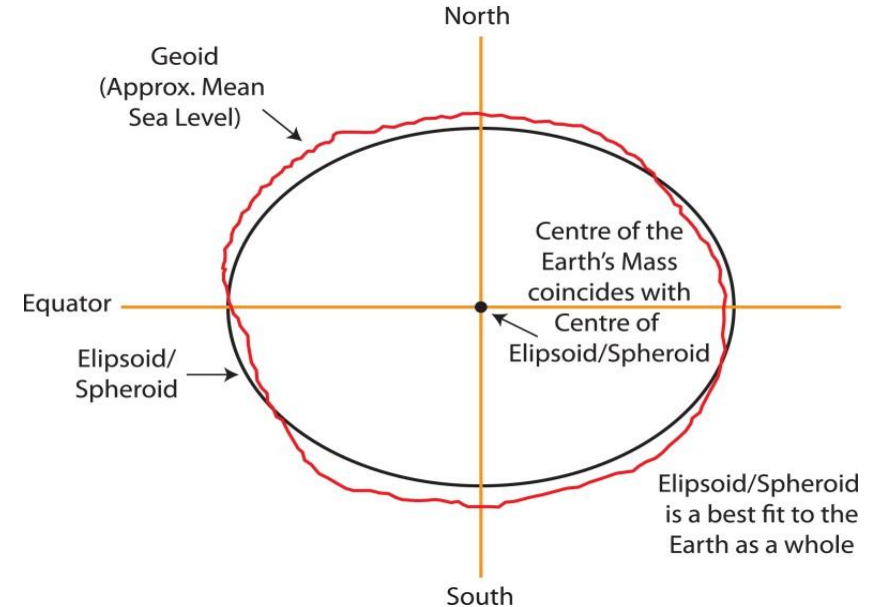
- Local or global
- Horizontal to measure positions (lat, lon) on the Earth or Vertical to measure the elevation of points on the Earth

A specific point on the Earth can have different coordinates depending on the datum used to make the measurement

Datums



North American Datum (NAD83)



World Geodetic System (WGS84)

source: <http://www.icsm.gov.au/mapping/datums1.html>

WGS is the default standard datum for coordinates today

SRID

The **Spatial Reference System Identifier** (SRID) identifies the coordinate system used to define spatial data

It is the primary key for the Open Geospatial Consortium (OGC) *spatial_ref_sys* metadata table for the Simple Features for SQL Specification. The SPATIAL_REF_SYS columns are:

- SRID: an integer value that uniquely identifies the Spatial Referencing System (SRS) within the database.
- NAME: The name of the standard or standards body that is being cited for this reference system. For example, "EPSG" would be a valid AUTH_NAME.
- AUTH_SRID: The ID of the Spatial Reference System as defined by the Authority cited in the AUTH_NAME. In the case of EPSG, this is where the EPSG projection code would go.
- SRTEXT: The Well-Known Text representation of the Spatial Reference System.

Source: <http://www.opengeospatial.org/standards/sfs> ,
<http://postgis.refractory.net/documentation/manual-1.4/ch04.html>

Representation modes for spatial data

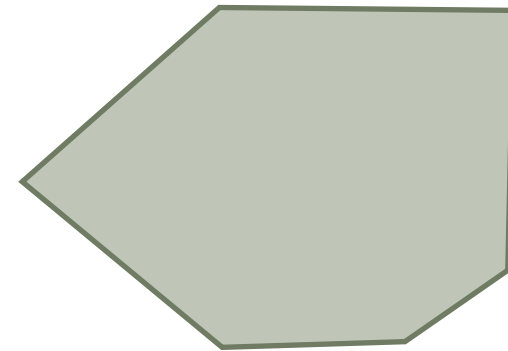
Tessellation: approximation of a continuous space by a discrete one

- a city is described by a set of cells that cover the city



Vector mode

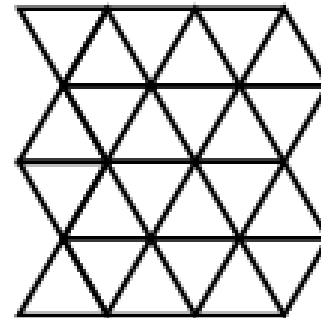
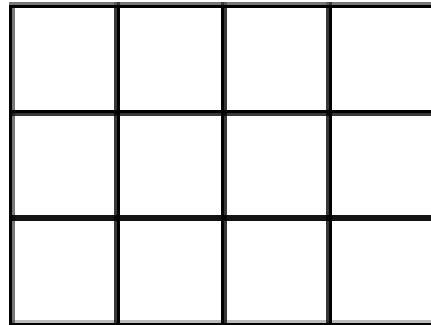
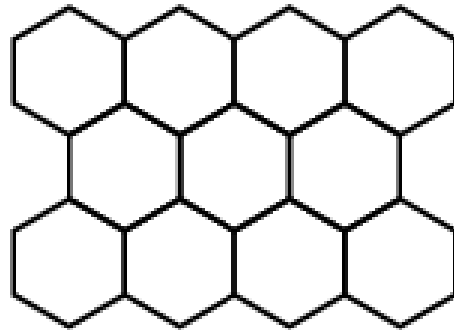
- a city is described by a list of points describing the boundary of a polygon



Tessellation

It is a discrete model also called spatial resolution model, tiling or mesh

Regular (fixed) tessellations: uses a regular grid or *raster* that is a collection of polygonal units of equal size



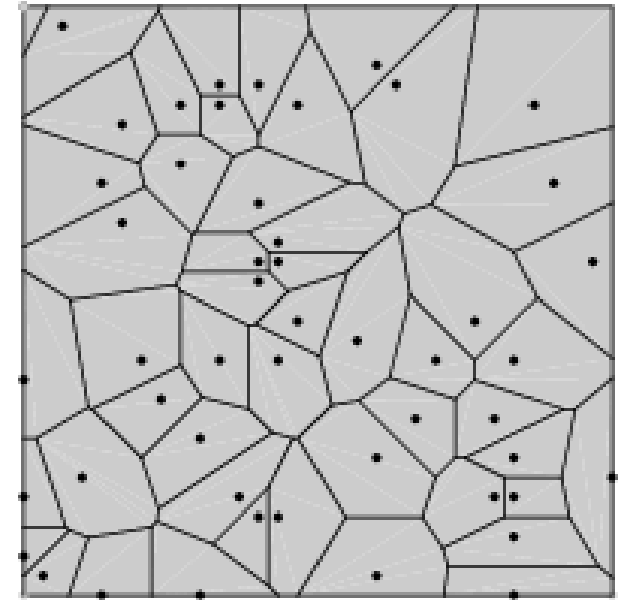
source: <http://mathworld.wolfram.com/RegularTessellation.html>

Tessellation

Irregular tessellations: the units of decomposition have different sizes

- The Thiessen polygons, also called Voronoi diagram, is a primer example of an irregular decomposition
- It is often used in zoning social, demographic or economical data

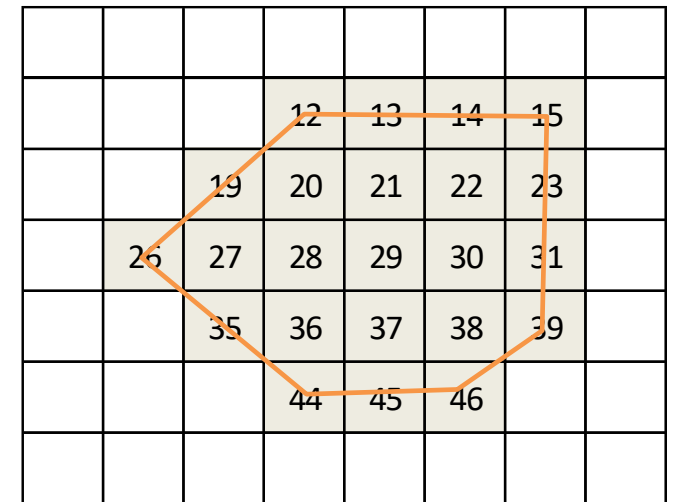
Given a set of points $\mathcal{P}$, the partitioning associates each point $p \in \mathcal{P}$ with a polygon that consists in the set of points whose closest point in $\mathcal{P}$ is p .



Tessellation

Discrete representation of a polygon

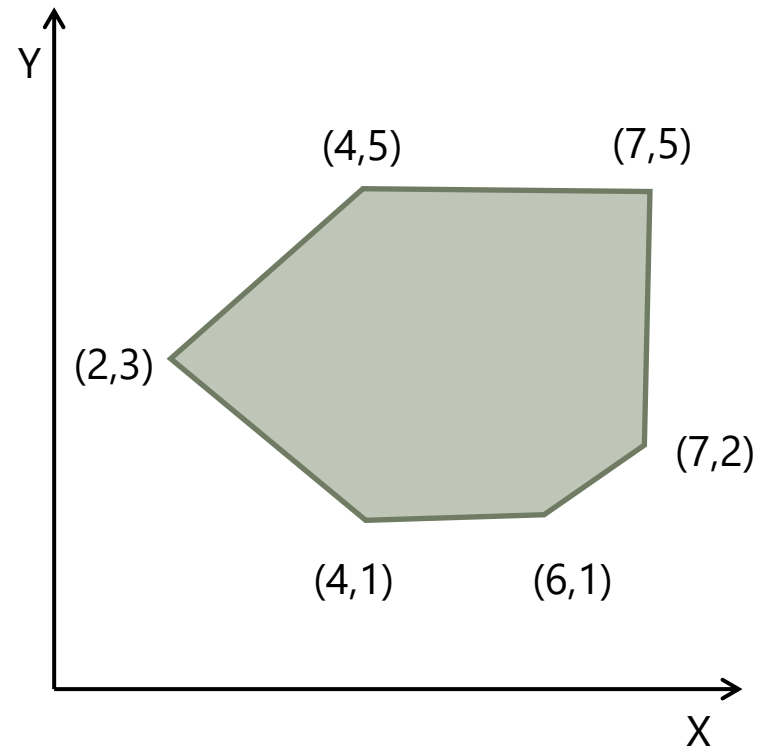
- a point is described by a single unit
- a polyline, a polygon or a region is represented by a finite set of units, e.g.:
< 12, 13, 14, 15, 19, 20, 21, 22, 23, 26, 27, 28, 29, 30, 31, 35, 36, 37, 38, 39, 44, 45, 46 >



Drawback: a faithful object representation occupies much memory space and operations are time consuming

Vector mode (continuous model)

- Objects are constructed from points and edges primitives
- Point (2D) is represented by a pair of coordinates
- Polyline, polygon or regions are represented by structures on the point representation (e.g., a polygon can be represented by a finite set of vertices)
- Memory requirements is usually lower than in raster representations



Software

Database

- PostgreSQL is available at <https://www.postgresql.org/download>
- PostGIS is available at <http://postgis.net/install/>

Do not forget to use the Spatial database template during install

You may also use *stackbuilder* (see *PostgreSQL tools just installed*) to install PostGIS

GIS

- Quantum GIS is available at <http://www.qgis.org/en/site/forusers/download.html>

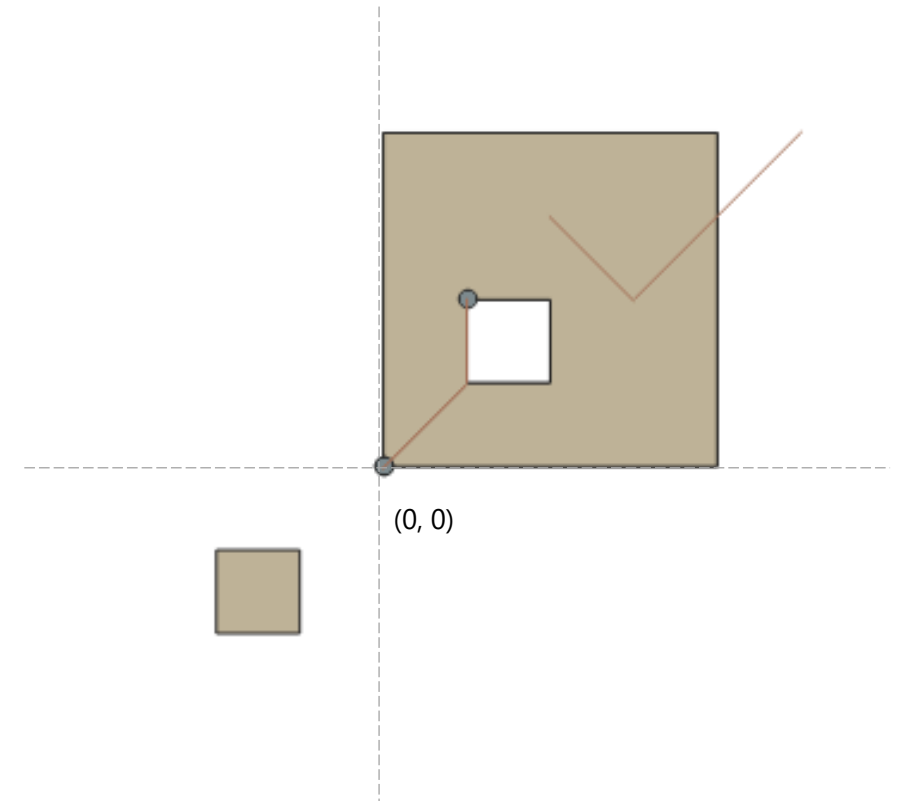
You will use the Desktop application

Geoserver

- Geoserver production is available at <http://geoserver.org/download/>

Spatial data types in WKT

- POINT(0 0)
- LINESTRING(0 0,1 1,1 2)
- POLYGON((0 0,4 0,4 4,0 4,0 0),(1 1,2 1,2 2,1 2,1 1))
- MULTIPOINT(0 0,1 2)
- MULTILINESTRING((0 0,1 1,1 2),(2 3,3 2,5 4))
- MULTIPOLYGON(((0 0,4 0,4 4,0 4,0 0),(1 1,2 1,2 2,1 2,1 1)), ((-1 -1,-1 -2,-2 -2,-2 -1,-1 -1)))
- GEOMETRYCOLLECTION(POINT(2 3),LINESTRING(2 3,3 4))



Bibliography

Nikos Mamoulis. Spatial data management. Synthesis Lectures on Data Management, Morgan&Claypool publishers, 2012.

- **Chapter 1**

contact

email: jose.moreira@ua.pt

Department of Electronics,
Telecommunications and Informatics
University of Aveiro
Portugal